

*Calculator*

**Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test.**

|                             |
|-----------------------------|
| <b>You have 85 minutes.</b> |
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Signature

**SHOW ALL WORK FOR FULL CREDIT!**

Show your set up (what you are putting into the calculator) and round your answers to the nearest thousandths.

Use a graphing calculator to find all solutions over the interval  $[0, 2\pi)$ . Express solutions to the thousandths. (2 points)

1.  $2x \cos x = 1$

2. Find the horizontal and vertical components for the vector  $\mathbf{v}$ , if it has the given direction angle and magnitude. Write the component form of vector  $\mathbf{v}$ . 3 pts

$$\alpha = 124^\circ 40', \quad |\mathbf{v}| = 9.8$$

*Calculator*

**SHOW ALL WORK FOR FULL CREDIT!**

Show your set up (what you are putting into the calculator) and round your answers to the nearest thousandths.

Two forces act at a point in the plane. The resultant force is given. Find the angle between the two forces. (4 points)

3. forces of 540 and 390 newtons, with a resultant force of 810 newtons

4. A force of 45 lb is required to hold a 160-lb crate on a ramp. What angle does the ramp make with the horizontal? 4 pts

*Calculator*

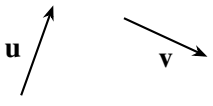
**SHOW ALL WORK FOR FULL CREDIT!**

Show your set up (what you are putting into the calculator) and round your answers to the nearest thousandths.

5. A pilot is flying at 235 mph. She wants her flight path to be on a bearing of  $48^{\circ}30'$ . A wind is blowing from the south at 24.5 mph. Find the bearing the pilot should fly, and find the plane's groundspeed. 4 pts

*Non-Calculator*

**SHOW ALL WORK FOR FULL CREDIT!**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| <p>6. Given vectors <math>\mathbf{u}</math> and <math>\mathbf{v}</math>, sketch and clearly label a single geometric diagram representing: <span style="float: right;">2 pts each</span></p> <div style="display: flex; justify-content: space-around; align-items: flex-start;"> <div style="text-align: center;">  <p><math>\mathbf{u}</math>      <math>\mathbf{v}</math></p> </div> <div style="text-align: center;">a) <math>\mathbf{u} - \mathbf{v}</math></div> <div style="text-align: center;">b) <math>\mathbf{u} + 2\mathbf{v}</math></div> </div> |
| <p>7. Given vectors <math>\mathbf{u}</math> and <math>\mathbf{v}</math>, find <span style="float: right;">5 pts</span></p> <p><math>\mathbf{u} = \langle 4, -1 \rangle</math>,    <math>\mathbf{v} = \langle -3, 6 \rangle</math></p> <div style="display: flex; justify-content: space-around;"> <div style="text-align: center;">a) <math>-4\mathbf{u}</math></div> <div style="text-align: center;">b) <math>2\mathbf{u} - 3\mathbf{v}</math></div> <div style="text-align: center;">c) <math>\mathbf{u} + \mathbf{v}</math></div> </div>                                                                                                   |
| <p>8. Find the magnitude and the positive direction angle for <math>\mathbf{u}</math>. <span style="float: right;">3 pts</span></p> <p><math>\mathbf{u} = \langle -5, -5 \rangle</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

9. For  $\mathbf{u} = \langle 6, -3 \rangle$  and  $\mathbf{v} = \langle -2, 7 \rangle$ , find  $\mathbf{u} \cdot \mathbf{v}$ .

2 pts

10. Determine whether the vectors are orthogonal.

2 pts

$$\mathbf{v} = 2\mathbf{i} + \sqrt{5}\mathbf{j}, \quad \mathbf{w} = \sqrt{5}\mathbf{i} - 2\mathbf{j}$$

Solve each equation algebraically on the given interval  $[0, 2\pi)$ . (3 points each)

11.  $2\cos 2\theta = 1$

12.  $2\sin^2 x = 2 + \cos x$

Solve the equation algebraically on the given interval  $[0^\circ, 360^\circ)$ . (3 points)

13.  $2 \sec^2 \theta - 3 \sec \theta = 2$

Solve the equation algebraically on the given interval  $[0^\circ, 360^\circ)$ . (4 points)

14.  $\sqrt{2} \csc \theta = 2 \cot \theta$

Find the exact value of each expression. (1 point each for 15 & 18 and 2 pts each for 16 & 17)

|                                                 |                                                 |
|-------------------------------------------------|-------------------------------------------------|
| 15. $\cos^{-1}\left(-\frac{\sqrt{2}}{2}\right)$ | 16. $\sin^{-1}\left(\sin \frac{4\pi}{3}\right)$ |
| 17. $\csc^{-1}\left(\sec \frac{\pi}{3}\right)$  | 18. $\sec^{-1}(-2)$                             |

Give the exact value of each expression. Rationalize the denominator of your answer. (3 points each)

|                                                            |                                                                                   |
|------------------------------------------------------------|-----------------------------------------------------------------------------------|
| 19. $\sec\left(2\cos^{-1}\left(-\frac{1}{4}\right)\right)$ | 20. $\cos\left(\tan^{-1}\frac{4}{3} + \sin^{-1}\left(-\frac{5}{13}\right)\right)$ |
|------------------------------------------------------------|-----------------------------------------------------------------------------------|

**ANSWER KEY – Calculator****SHOW ALL WORK FOR FULL CREDIT!**

Show your set up (what you are putting into the calculator) and round your answers to the nearest thousandths.

Use a graphing calculator to find all solutions over the interval  $[0, 2\pi)$ . Express solutions to the thousandths. (2 points)

1.  $2x \cos x = 1$

Cannot be solved algebraically ( $x$  is both inside and outside the trig function). Set up the calculator, radian mode:

$$Y_1 = 2x \cos x - 1 \quad \text{window } [0, 2\pi) \text{ by } [-6, 14]$$

menu  $\rightarrow$  Analyze Graph  $\rightarrow$  Zero (or graph  $Y_1 = 2x \cos x$ ,  $Y_2 = 1$  and find the intersections)

Three zeros in the interval. The curve dips below 1 after  $x \approx 1.1$  and climbs back through 1 near the right end, so check all the way to  $2\pi$ .

$$x \approx 0.610, 1.098, 4.816$$

2. Find the horizontal and vertical components for the vector  $\mathbf{v}$ , if it has the given direction angle and magnitude. Write the component form of vector  $\mathbf{v}$ . 3 pts

$$\alpha = 124^\circ 40', \quad |\mathbf{v}| = 9.8$$

horizontal:  $a = |\mathbf{v}| \cos \alpha$  vertical:  $b = |\mathbf{v}| \sin \alpha$  ( $124^\circ 40' = 124^\circ + \frac{40}{60}^\circ = 124.6^\circ$ , or type the DMS straight in)

$$a = 9.8 \cos(124^\circ 40') \approx -5.574$$

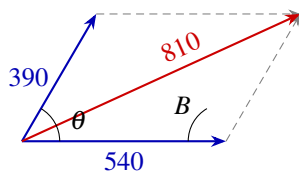
$$b = 9.8 \sin(124^\circ 40') \approx 8.060$$

$$\mathbf{v} \approx \langle -5.574, 8.060 \rangle$$

Signs check:  $124^\circ 40'$  is in QII, so  $a < 0$  and  $b > 0$ .

Two forces act at a point in the plane. The resultant force is given. Find the angle between the two forces. (4 points)

3. forces of 540 and 390 newtons, with a resultant force of 810 newtons



Same parallelogram, but now the resultant is known and the angle is not. Call the angle across from the resultant  $B$ ; the angle between the forces is its supplement,  $\theta = 180^\circ - B$ . Law of Cosines with the resultant as the side across from  $B$ , solved for  $\cos B$ .

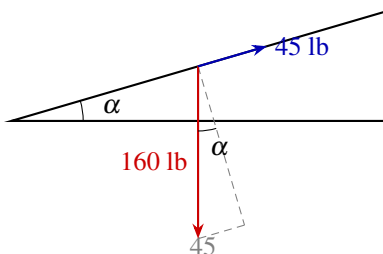
$$810^2 = 540^2 + 390^2 - 2(540)(390)\cos B \quad \Rightarrow \quad 656100 = 443700 - 421200\cos B$$

$$\cos B = \frac{656100 - 443700}{-421200} \approx -0.50427 \quad \Rightarrow \quad B \approx 120.283^\circ \quad \Rightarrow \quad \theta = 180^\circ - 120.283^\circ \approx 59.717^\circ$$

The angle between the two forces is about  $59.717^\circ$ . (Sanity check: 810 is well under  $540 + 390 = 930$ , so the forces are spread apart, not nearly parallel.)



4. A force of 45 lb is required to hold a 160-lb crate on a ramp. What angle does the ramp make with the horizontal? 4 pts

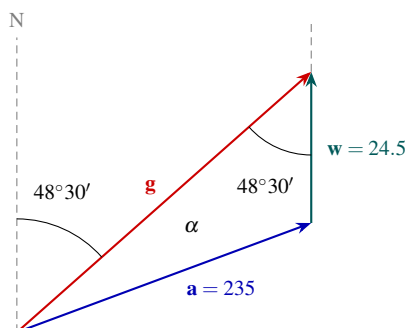


The weight (160 lb, straight down) is the hypotenuse. The 45 lb holding force balances the component of the weight parallel to the ramp, which is the leg across from the ramp angle  $\alpha$  (the angle between the weight and the perpendicular to the ramp equals the ramp angle). SOH, then  $\sin^{-1}$ :

$$\sin \alpha = \frac{45}{160} = 0.28125 \implies \alpha = \sin^{-1}(0.28125) \approx \boxed{16.335^\circ}$$

The ramp makes an angle of about  $16.335^\circ$  with the horizontal.

5. A pilot is flying at 235 mph. She wants her flight path to be on a bearing of  $48^\circ 30'$ . A wind is blowing from the south at 24.5 mph. Find the bearing the pilot should fly, and find the plane's groundspeed. 4 pts



$\mathbf{a} + \mathbf{w} = \mathbf{g}$ . Draw what you know best first: the course  $\mathbf{g}$  on a bearing of  $48^\circ 30'$ , and the wind *from* the south, so  $\mathbf{w}$  points due north (bearing  $0^\circ$ ) with its tip at the tip of  $\mathbf{g}$ . The heading of  $\mathbf{a}$  is the unknown. At the tip of  $\mathbf{g}$ , the north line and  $\mathbf{g}$  make the same  $48^\circ 30'$  (alternate interior angles), and that angle sits across from  $\mathbf{a} = 235$ . Law of Sines for the drift angle  $\alpha$  (across from  $\mathbf{w}$ ), then for  $|\mathbf{g}|$ .

not to scale

$$\frac{\sin \alpha}{24.5} = \frac{\sin 48^\circ 30'}{235} \implies \sin \alpha \approx 0.07808 \implies \alpha \approx 4.478^\circ$$

The wind pushes the plane north (counterclockwise off the course), so she aims clockwise of the course:

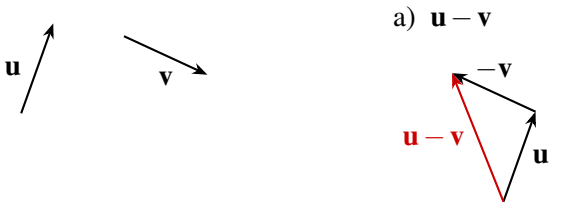
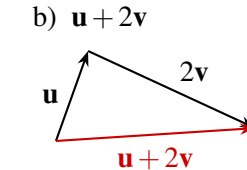
$$\text{heading} = 48^\circ 30' + 4.478^\circ \approx \boxed{52.978^\circ} \quad (\approx 52^\circ 58' 42'')$$

Third angle =  $180^\circ - 48.5^\circ - 4.478^\circ = 127.022^\circ$ , across from  $\mathbf{g}$ :

$$\frac{|\mathbf{g}|}{\sin 127.022^\circ} = \frac{235}{\sin 48^\circ 30'} \implies |\mathbf{g}| \approx \boxed{250.517 \text{ mph}}$$

The pilot should fly on a bearing of about  $52.978^\circ$ , and the plane's groundspeed is about 250.517 mph.

**ANSWER KEY – Non-Calculator****SHOW ALL WORK FOR FULL CREDIT!**

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| <p>6. Given vectors <math>\mathbf{u}</math> and <math>\mathbf{v}</math>, sketch and clearly label a single geometric diagram representing: <span style="float: right;">2 pts each</span></p> <div style="display: flex; justify-content: space-around; align-items: flex-start;"> <div style="text-align: center;">  <p>a) <math>\mathbf{u} - \mathbf{v}</math></p> <p>Subtracting is adding the opposite: put <math>-\mathbf{v}</math> (same length as <math>\mathbf{v}</math>, reversed) tail on the tip of <math>\mathbf{u}</math>. The difference runs from the tail of <math>\mathbf{u}</math> to the tip of <math>-\mathbf{v}</math>.</p> </div> <div style="text-align: center;">  <p>b) <math>\mathbf{u} + 2\mathbf{v}</math></p> <p>Double <math>\mathbf{v}</math> (same direction, twice the length) and hang it off the tip of <math>\mathbf{u}</math>; the sum closes the triangle from the tail of <math>\mathbf{u}</math>.</p> </div> </div> |  |
| <p>7. Given vectors <math>\mathbf{u}</math> and <math>\mathbf{v}</math>, find <span style="float: right;">5 pts</span></p> <p><math>\mathbf{u} = \langle 4, -1 \rangle, \quad \mathbf{v} = \langle -3, 6 \rangle</math></p> <p>a) <math>-4\mathbf{u}</math> <span style="margin-left: 100px;">b) <math>2\mathbf{u} - 3\mathbf{v}</math></span> <span style="margin-left: 100px;">c) <math>\mathbf{u} + \mathbf{v}</math></span></p> <p>Given in component form, so answer in component form.</p> <div style="display: flex; justify-content: space-between;"> <div> <math>-4\langle 4, -1 \rangle = \boxed{\langle -16, 4 \rangle}</math> </div> <div> <math>2\langle 4, -1 \rangle - 3\langle -3, 6 \rangle = \langle 8, -2 \rangle - \langle -9, 18 \rangle</math><br/> <math>= \langle 8 + 9, -2 - 18 \rangle = \boxed{\langle 17, -20 \rangle}</math> </div> <div> <math>\langle 4 + (-3), -1 + 6 \rangle</math><br/> <math>= \boxed{\langle 1, 5 \rangle}</math> </div> </div>                                                                                                                                           |  |
| <p>8. Find the magnitude and the positive direction angle for <math>\mathbf{u}</math>. <span style="float: right;">3 pts</span></p> <p><math>\mathbf{u} = \langle -5, -5 \rangle</math></p> <div style="display: flex; justify-content: space-around; align-items: center;"> <div> <math> \mathbf{u}  = \sqrt{(-5)^2 + (-5)^2} = \sqrt{50} = \boxed{5\sqrt{2}}</math> </div> <div> <math>\tan \theta = \frac{b}{a} = \frac{-5}{-5} = 1 \implies \text{reference angle } 45^\circ</math> </div> </div> <p><math>a &lt; 0, b &lt; 0</math> puts <math>\mathbf{u}</math> in QIII (tangent is positive there, which is why <math>\tan^{-1}(1) = 45^\circ</math> alone is wrong), so <math>\theta = 180^\circ + 45^\circ = \boxed{225^\circ}</math> or <math>\boxed{\theta = \frac{5\pi}{4}}</math>.</p> <p>The direction angle comes from the quadrant of the vector, never straight off <math>\tan^{-1}</math>.</p>                                                                                                                                                                                                              |  |
| <p>9. For <math>\mathbf{u} = \langle 6, -3 \rangle</math> and <math>\mathbf{v} = \langle -2, 7 \rangle</math>, find <math>\mathbf{u} \cdot \mathbf{v}</math>. <span style="float: right;">2 pts</span></p> <p style="text-align: center;"><math>\mathbf{u} \cdot \mathbf{v} = (6)(-2) + (-3)(7) = -12 - 21 = \boxed{-33}</math></p> <p>A scalar, not a vector. Negative, so the angle between them is obtuse.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
| <p>10. Determine whether the vectors are orthogonal. <span style="float: right;">2 pts</span></p> <p><math>\mathbf{v} = 2\mathbf{i} + \sqrt{5}\mathbf{j}, \quad \mathbf{w} = \sqrt{5}\mathbf{i} - 2\mathbf{j}</math></p> <p>Orthogonal means the dot product is 0. <math>\mathbf{v} = \langle 2, \sqrt{5} \rangle, \mathbf{w} = \langle \sqrt{5}, -2 \rangle</math>.</p> <p style="text-align: center;"><math>\mathbf{v} \cdot \mathbf{w} = (2)(\sqrt{5}) + (\sqrt{5})(-2) = 2\sqrt{5} - 2\sqrt{5} = 0</math> <span style="border: 1px solid black; padding: 2px 10px;">Yes, orthogonal</span></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |

Solve each equation algebraically on the given interval  $[0, 2\pi)$ . (3 points each)

11.  $2\cos 2\theta = 1$

Isolate the trig function, then solve for the whole argument  $2\theta$  first.

$$\cos 2\theta = \frac{1}{2}$$

Cosine positive: QI, QIV; reference angle  $\frac{\pi}{3}$ . General solution on  $2\theta$ :

$$2\theta = \frac{\pi}{3} + 2\pi k \quad \text{or} \quad 2\theta = \frac{5\pi}{3} + 2\pi k$$

Divide everything by 2, the  $k$  term included:

$$\theta = \frac{\pi}{6} + \pi k \quad \text{or} \quad \theta = \frac{5\pi}{6} + \pi k$$

$$k = 0: \frac{\pi}{6}, \frac{5\pi}{6} \quad k = 1: \frac{7\pi}{6}, \frac{11\pi}{6} \quad k = 2: \text{too big.}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

12.  $2\sin^2 x = 2 + \cos x$

Two different functions: use  $\sin^2 x = 1 - \cos^2 x$  so everything is in  $\cos x$ .

$$2(1 - \cos^2 x) = 2 + \cos x$$

$$2 - 2\cos^2 x - 2 - \cos x = 0 \implies -2\cos^2 x - \cos x = 0$$

Factor the GCF (do not divide by  $\cos x$ ):

$$\cos x (2\cos x + 1) = 0$$

$$\cos x = 0 \implies x = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$\cos x = -\frac{1}{2} \implies x = \frac{2\pi}{3}, \frac{4\pi}{3}$$

$$x = \frac{\pi}{2}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{3\pi}{2}$$

Solve the equation algebraically on the given interval  $[0^\circ, 360^\circ)$ . (3 points)

13.  $2\sec^2 \theta - 3\sec \theta = 2$

One trig function, quadratic in form. Set equal to 0 and factor.

$$2\sec^2 \theta - 3\sec \theta - 2 = 0 \implies (2\sec \theta + 1)(\sec \theta - 2) = 0$$

$$\sec \theta = -\frac{1}{2} \implies \cos \theta = -2 \quad \text{impossible } (|\cos \theta| \leq 1), \text{ no solutions from this factor}$$

$$\sec \theta = 2 \implies \cos \theta = \frac{1}{2} \implies \theta = 60^\circ, 300^\circ \text{ (QI, QIV)}$$

$$\theta = 60^\circ, 300^\circ$$

Solve the equation algebraically on the given interval  $[0^\circ, 360^\circ)$ . (4 points)

14.  $\sqrt{2} \csc \theta = 2 \cot \theta$

Two different functions and no square to convert, so rewrite both in sine and cosine:  $\csc \theta = \frac{1}{\sin \theta}$ ,  $\cot \theta = \frac{\cos \theta}{\sin \theta}$ .

$$\frac{\sqrt{2}}{\sin \theta} = \frac{2\cos \theta}{\sin \theta}$$

Both sides already require  $\sin \theta \neq 0$  (or  $\csc \theta$  and  $\cot \theta$  would be undefined), so multiplying through by  $\sin \theta$  loses nothing here:

$$\sqrt{2} = 2\cos \theta \implies \cos \theta = \frac{\sqrt{2}}{2} \implies \theta = 45^\circ, 315^\circ$$

$$\text{Check } 45^\circ: \sqrt{2} \cdot \sqrt{2} = 2 \text{ and } 2 \cot 45^\circ = 2. \checkmark$$

$$\theta = 45^\circ, 315^\circ$$

Find the exact value of each expression. (1 point each for 15 & 18 and 2 pts each for 16 & 17)

|                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <p>15. <math>\cos^{-1}\left(-\frac{\sqrt{2}}{2}\right)</math></p> <p>Range of <math>\cos^{-1}</math> is <math>[0, \pi]</math> (QI, QII). Negative input goes to QII, reference angle <math>\frac{\pi}{4}</math>.</p> <div style="border: 1px solid black; padding: 5px; width: fit-content; margin: 10px auto;"> <math>\frac{3\pi}{4}</math> </div>                                                        | <p>16. <math>\sin^{-1}\left(\sin \frac{4\pi}{3}\right)</math></p> <p><math>\frac{4\pi}{3}</math> is not in <math>[-\frac{\pi}{2}, \frac{\pi}{2}]</math>, so it cannot be the answer. Inside first:</p> $\sin \frac{4\pi}{3} = -\frac{\sqrt{3}}{2} \implies \sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$ <p>Range <math>[-\frac{\pi}{2}, \frac{\pi}{2}]</math>, negative sine means QIV: <div style="border: 1px solid black; padding: 5px; width: fit-content; margin: 10px auto;"><math>-\frac{\pi}{3}</math></div></p> |
| <p>17. <math>\csc^{-1}\left(\sec \frac{\pi}{3}\right)</math></p> <p>Inside first: <math>\sec \frac{\pi}{3} = \frac{1}{\cos \frac{\pi}{3}} = \frac{1}{1/2} = 2</math>.</p> $\csc^{-1}(2) = \theta \implies \csc \theta = 2 \implies \sin \theta = \frac{1}{2} \implies \frac{\pi}{6}$ <p>(range of <math>\csc^{-1}</math> matches <math>\sin^{-1}</math> with 0 excluded; a positive input lands in QI)</p> | <p>18. <math>\sec^{-1}(-2)</math></p> <p>Range of <math>\sec^{-1}</math> is <math>[0, \pi]</math> with <math>\frac{\pi}{2}</math> excluded (QI, QII), so a negative input goes to QII.</p> $\sec \theta = -2 \implies \cos \theta = -\frac{1}{2}$ $\theta = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$ <div style="border: 1px solid black; padding: 5px; width: fit-content; margin: 10px auto;"><math>\frac{2\pi}{3}</math></div>                                                                                         |

Give the exact value of each expression. Rationalize the denominator of your answer. (3 points each)

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <p>19. <math>\sec\left(2\cos^{-1}\left(-\frac{1}{4}\right)\right)</math></p> <p>Let <math>u = \cos^{-1}\left(-\frac{1}{4}\right)</math>, so <math>\cos u = -\frac{1}{4}</math> (QII). Pick the double-angle form that only needs cosine:</p> $\begin{aligned}\cos 2u &= 2\cos^2 u - 1 = 2\left(-\frac{1}{4}\right)^2 - 1 \\ &= 2\left(\frac{1}{16}\right) - 1 = \frac{1}{8} - 1 = -\frac{7}{8}\end{aligned}$ $\sec 2u = \frac{1}{\cos 2u} = \frac{1}{-\frac{7}{8}} = -\frac{8}{7}$ <p>(No triangle needed. With <math>1 - 2\sin^2 u</math> instead: <math>x = -1</math>, <math>r = 4</math>, <math>y = \sqrt{15}</math>, <math>\sin u = \frac{\sqrt{15}}{4}</math>, and <math>1 - 2 \cdot \frac{15}{16} = -\frac{7}{8}</math>. Same answer.)</p> | <p>20. <math>\cos\left(\tan^{-1} \frac{4}{3} + \sin^{-1}\left(-\frac{5}{13}\right)\right)</math></p> <p><math>\cos(A+B) = \cos A \cos B - \sin A \sin B</math>.</p> <p><math>A = \tan^{-1} \frac{4}{3}</math>: <math>x = 3</math>, <math>y = 4</math>, <math>r = 5</math> (QI), so <math>\cos A = \frac{3}{5}</math>, <math>\sin A = \frac{4}{5}</math>.</p> <p><math>B = \sin^{-1}\left(-\frac{5}{13}\right)</math>: <math>y = -5</math>, <math>r = 13</math>, <math>x = 12</math> (QIV, from the range of <math>\sin^{-1}</math>), so <math>\cos B = \frac{12}{13}</math>, <math>\sin B = -\frac{5}{13}</math>.</p> $\begin{aligned}\left(\frac{3}{5}\right)\left(\frac{12}{13}\right) - \left(\frac{4}{5}\right)\left(-\frac{5}{13}\right) \\ = \frac{36}{65} + \frac{20}{65} = \frac{56}{65}\end{aligned}$ <p>Nothing to rationalize; the denominator is already rational.</p> |
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